Table Connections by Module

1. RBAC Module (Admin / Roles / Login)

Tables: `GPO_ROLE_MASTER`, `GPO_ROLE_PERMISSION`, `GPO_ROLE_MAPPING`, `GPO_USER_LOGIN`

Child Table	FK Column	Parent Table	Parent Column
`GPO_ROLE_PERMISSION`	`ROLE_ID`	`GPO_ROLE_MASTER`	`ROLE_ID`
`GPO_ROLE_MAPPING`	`ROLE_ID`	`GPO_ROLE_MASTER`	`ROLE_ID`
`GPO_USER_LOGIN`	`USER_ID`	`GPO_ROLE_MAPPING`	`USER_ID`

Flow: `GPO_ROLE_MASTER` is the hub — it defines a role and its CRUD permissions (`GPO_ROLE_PERMISSION`) and which users are assigned that role (`GPO_ROLE_MAPPING`). Each mapped user's login/token sessions are tracked in `GPO_USER_LOGIN`, linked back to `GPO_ROLE_MAPPING` via `USER_ID`.

```
GPO_ROLE_MASTER ─┬─→ GPO_ROLE_PERMISSION   (ROLE_ID)
                 └─→ GPO_ROLE_MAPPING      (ROLE_ID)
                          └─→ GPO_USER_LOGIN   (USER_ID)
```

2. Contract / Membership Module

Tables: `MN_MEMBER`, `MN_ORGANIZATION_ID`, `MN_CF_ELIGIBLE_CUSTOMER`, `MN_STRUCTURED_DOC`, `MN_STRUCTURED_CONTRACT`

Child Table	FK Column	Parent Table	Parent Column
`MN_ORGANIZATION_ID`	`MEMBER_ID`	`MN_MEMBER`	`MEMBER_ID`
`MN_CF_ELIGIBLE_CUSTOMER`	`MEMBER_ID`	`MN_MEMBER`	`MEMBER_ID`
`MN_STRUCTURED_CONTRACT`	`STRUCT_DOC_ID`	`MN_STRUCTURED_DOC`	`STRUCT_DOC_ID`

Flow: `MN_MEMBER` is the hub for membership data — one member can have multiple organization IDs and multiple eligible-customer records. Separately, `MN_STRUCTURED_DOC` is the hub for contract documents — one document can have multiple structured contracts.

```
MN_MEMBER ─┬─→ MN_ORGANIZATION_ID          (MEMBER_ID)
           └─→ MN_CF_ELIGIBLE_CUSTOMER     (MEMBER_ID)

MN_STRUCTURED_DOC ──→ MN_STRUCTURED_CONTRACT   (STRUCT_DOC_ID)
```

3. Upload Module

Tables: `GPO_INPUT_FORMAT`, `GPO_STAGED_INPUT`, `GPO_CUST_INPUT`, `GPO_CUST_HDR`

Flow: A file upload creates a `GPO_CUST_HDR` record (batch header) and inserts raw rows into `GPO_STAGED_INPUT`, keyed by `batch_num`. `GPO_INPUT_FORMAT` stores the column-mapping template (`format_name`) used to map staged columns into the standardized `GPO_CUST_INPUT` table, also keyed by `batch_num`.

```
GPO_CUST_HDR (batch_num)
   └──> GPO_STAGED_INPUT     (batch_num)
          └──> GPO_CUST_INPUT   (batch_num, mapped using GPO_INPUT_FORMAT)
```

4. HIN Validation Module

Tables: `GPO_CUST_INPUT`, `GPO_HIN_VALIDATE`, `GPO_HIN`, `GPO_CUST_HDR`, `GPO_LOGS`

Flow: Records from `GPO_CUST_INPUT` (filtered by `batch_num`) are matched against the master `GPO_HIN` lookup table by `hin`/name. Results are written to `GPO_HIN_VALIDATE`, keyed by `batch_num`. The batch's `hin_validated` flag on `GPO_CUST_HDR` is updated, and the action is recorded in `GPO_LOGS`.

```
GPO_CUST_INPUT (batch_num) ──compare──> GPO_HIN (master lookup)
        └──> GPO_HIN_VALIDATE (batch_num)
GPO_CUST_HDR.hin_validated ← updated
GPO_LOGS ← action recorded
```

5. DEA Validation Module

Tables: `GPO_CUST_INPUT`, `GPO_DEA_VALIDATE`, `GPO_DEA`, `GPO_CUST_HDR`, `GPO_LOGS`

Flow: Same pattern as HIN — `GPO_CUST_INPUT` records are checked against the `GPO_DEA` master table, results stored in `GPO_DEA_VALIDATE` by `batch_num`, and `GPO_CUST_HDR.dea_validated` is updated with a `GPO_LOGS` entry.

```
GPO_CUST_INPUT (batch_num) ──compare──> GPO_DEA (master lookup)
        └──> GPO_DEA_VALIDATE (batch_num)
GPO_CUST_HDR.dea_validated ← updated
GPO_LOGS ← action recorded
```

6. HRSA Validation Module

Tables: `GPO_CUST_INPUT`, `GPO_HRSA_VALIDATE`, `GPO_HRSA`, `GPO_CUST_HDR`, `GPO_LOGS`

Flow: `GPO_CUST_INPUT` records are checked against the `GPO_HRSA` master table, results stored in `GPO_HRSA_VALIDATE` by `batch_num`, and `GPO_CUST_HDR.hrsa_validated` is updated with a `GPO_LOGS` entry.

```
GPO_CUST_INPUT (batch_num) ──compare──> GPO_HRSA (master lookup)
        └─> GPO_HRSA_VALIDATE (batch_num)
GPO_CUST_HDR.hrsa_validated ← updated
GPO_LOGS ← action recorded
```

7. USPS (Address) Validation Module

Tables: `GPO_CUST_INPUT`, `GPO_USPS_VALIDATE`, `GPO_USPS`, `GPO_CUST_HDR`, `GPO_LOGS`

Flow: Address fields (`street`, `city`, `state`, `postal_code`) from `GPO_CUST_INPUT` are checked/corrected against the `GPO_USPS` master table, with results in `GPO_USPS_VALIDATE` by `batch_num`. `GPO_CUST_HDR.usps_validated` is updated and logged in `GPO_LOGS`.

```
GPO_CUST_INPUT (batch_num) ──compare──> GPO_USPS (master lookup)
        └─> GPO_USPS_VALIDATE (batch_num)
GPO_CUST_HDR.usps_validated ← updated
GPO_LOGS ← action recorded
```

8. COT (Class of Trade) Validation Module

Tables: `GPO_CUST_INPUT`, `GPO_COT_VALIDATE`, `GPO_CUST_HDR`, `GPO_LOGS`

Flow: COT values from `GPO_CUST_INPUT` are cross-checked against HIN/DEA validation outcomes (`hibcc_cot`, `dea_cot`) to produce a corrected COT in `GPO_COT_VALIDATE`, keyed by `batch_num`/`customer_name`. `GPO_CUST_HDR.cot_validated` is updated and logged.

```
GPO_CUST_INPUT (batch_num) + GPO_HIN_VALIDATE/GPO_DEA_VALIDATE (cot fields)
        └─> GPO_COT_VALIDATE (batch_num, customer_name)
GPO_CUST_HDR.cot_validated ← updated
GPO_LOGS ← action recorded
```

9. Member Validation Module

Tables: `GPO_CUST_INPUT`, `GPO_MEMBER_VALIDATE`, `GPO_CUST_HDR`, `GPO_LOGS`

Flow: `GPO_CUST_INPUT` records are validated for identifier/address completeness and membership match criteria, with results written to `GPO_MEMBER_VALIDATE` by `BATCH_NUM`/`NAME`. `GPO_CUST_HDR.member_validated` is updated and logged.

```
GPO_CUST_INPUT (BATCH_NUM)
        └──> GPO_MEMBER_VALIDATE (BATCH_NUM, NAME)
GPO_CUST_HDR.member_validated ← updated
GPO_LOGS ← action recorded
```

10. Staging Output Module

Tables: `GPO_CUST_INPUT`, `GPO_HIN_VALIDATE`, `GPO_DEA_VALIDATE`, `GPO_USPS_VALIDATE`, `GPO_HRSA_VALIDATE`, `GPO_COT_VALIDATE`, `GPO_MEMBER_VALIDATE`, `GPO_CUST_STG_OUTPUT`, `GPO_LOGS`

Flow: This module merges all validation results together. For each `GPO_CUST_INPUT` record in a batch, it looks up matching rows (by `batch_num` + name/hin/dea/street/etc.) across `GPO_HIN_VALIDATE`, `GPO_DEA_VALIDATE`, `GPO_USPS_VALIDATE`, `GPO_HRSA_VALIDATE`, `GPO_COT_VALIDATE`, and `GPO_MEMBER_VALIDATE`, then writes the combined, corrected record into `GPO_CUST_STG_OUTPUT`. The action is logged in `GPO_LOGS`.

```
GPO_CUST_INPUT (batch_num)
   ├──> GPO_HIN_VALIDATE     (batch_num, original_hin)
   ├──> GPO_DEA_VALIDATE     (batch_num, original_dea)
   ├──> GPO_USPS_VALIDATE    (batch_num, original_street/city/state/postal_code)
   ├──> GPO_HRSA_VALIDATE    (batch_num, original_hrsa)
   ├──> GPO_COT_VALIDATE     (batch_num, customer_name)
   └──> GPO_MEMBER_VALIDATE  (BATCH_NUM, NAME)
            all merged into
        GPO_CUST_STG_OUTPUT (batch_num, name)
GPO_LOGS ← action recorded
```

11. Final Output Module

Tables: `GPO_CUST_STG_OUTPUT`, `GPO_CUST_OUTPUT`, `GPO_OUTPUT_FORMAT`, `GPO_CUST_HDR`, `GPO_LOGS`

Flow: Staged/merged records in `GPO_CUST_STG_OUTPUT` are transformed into the final `GPO_CUST_OUTPUT` table for the batch, using a chosen `GPO_OUTPUT_FORMAT` template. `GPO_CUST_HDR.isOutputGenerated` is updated and logged in `GPO_LOGS`.

```
GPO_CUST_STG_OUTPUT (batch_num) ──format via GPO_OUTPUT_FORMAT──> GPO_CUST_OUTPUT (ba
GPO_CUST_HDR.isOutputGenerated ← updated
GPO_LOGS ← action recorded
```

12. Session / Validation Job Tracking Module

Tables: `GPO_CUST_HDR`, `GPO_CUST_INPUT`, `GPO_VALIDATION_JOBS`, `GPO_LOGS`

Flow: Every batch header (`GPO_CUST_HDR`) has a corresponding `GPO_VALIDATION_JOBS` row (same `batch_num`) tracking the status of each stage — `member_status`, `hin_status`, `dea_status`, `usps_status`, `cot_status`, `hrsa_status`, `stg_status`, `output_status`. The session controller reads

`GPO_CUST_HDR.CURRENT_ACTION` to decide which stage's data (from `GPO_CUST_INPUT` or the relevant validation table) to return next, and every state change is written to `GPO_LOGS`.

```
GPO_CUST_HDR (batch_num, CURRENT_ACTION)
   └─→ GPO_VALIDATION_JOBS (batch_num) — per-stage status
   └─→ GPO_CUST_INPUT (batch_num) — data source per stage
GPO_LOGS ← every action recorded
```

13. Execution History Module

Tables: `GPO_CUST_HDR`

Flow: Reuses `GPO_CUST_HDR` directly (same table, mapped via a separate `ExecutionHistory` model) to list past batches with their validation flags (`HIN_VALIDATED`, `DEA_VALIDATED`, `USPS_VALIDATED`, `COT_VALIDATED`, `HRSA_VALIDATED`, `isOutputGenerated`) for history/reporting views.

```
GPO_CUST_HDR (batch_num) → read-only history view
```

14. Logs Module

Tables: `GPO_LOGS`

Flow: Central audit table. Every other module (upload, HIN, DEA, HRSA, USPS, COT, member, staging, output, session) writes a row here keyed by `batch_num`, recording `user_action`, `action_desc`, and who/when.

```
[All modules] ──insert──→ GPO_LOGS (batch_num, user_action, action_desc)
```

Overall Pipeline Connection (all modules combined)

```
GPO_CUST_HDR (batch_num)
   ├─→ GPO_CUST_INPUT
   │      └─→ GPO_STAGED_INPUT (via GPO_INPUT_FORMAT mapping)
   │
   ├─→ GPO_MEMBER_VALIDATE
   ├─→ GPO_HIN_VALIDATE   ──ref── GPO_HIN
   ├─→ GPO_DEA_VALIDATE   ──ref── GPO_DEA
   ├─→ GPO_USPS_VALIDATE  ──ref── GPO_USPS
   ├─→ GPO_HRSA_VALIDATE  ──ref── GPO_HRSA
   ├─→ GPO_COT_VALIDATE
   │
   ├─→ GPO_CUST_STG_OUTPUT (merges all validation results)
   │      └─→ GPO_CUST_OUTPUT (via GPO_OUTPUT_FORMAT)
   │
```

```
    ├──→ GPO_VALIDATION_JOBS (per-stage status)
    └──→ GPO_LOGS (audit trail)
```